Topic II: The Marchenko-Pastur Law

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We now discuss PCA in high dimensions, meaning when p, the dimensionality of the data, is comparable in size (or even larger than) n, the number of observations. To formalize this, we consider a sequence of observations models where both the number of observations and the dimensionality grow; that is, where $p \to \infty$. For simplicity, we assume our n iid observations $X_1, \dots, X_n$ in $\mathbb{R}^p$ have mean $\mathbf{0}$ and covariance $\mathbf{I}_p$, and define the sample covariance

$$\widehat{\mathbf{\Sigma}}_n = \frac{1}{n} \sum_{j=1}^{n} X_j X_j^T. \tag{1}$$

Of course, if p is fixed, Weyl's Theorem or Wielandt-Hoffman show that each of the p eigenvalues of the sample covariance converges to 1 as $n \to \infty$. However, when $p = p(n)$ grows with n, an entirely different phenomenon emerges that we will describe, and prove.

Since both p and n are growing to infinity, we need to impose some additional constraints on their relative sizes. We will suppose that $p = p(n)$ is a function of n, and that there is a number $\gamma \in (0, \infty)$ such that

$$\lim_{n\to\infty} \frac{p(n)}{n} = \gamma. \tag{2}$$

1 The Marchenko-Pastur Law

We consider iid random vectors in $\mathbb{R}^p$, $X_1, \dots, X_n \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_p)$; note that (a) we are assuming the mean is $\mathbf{0}$; and (b) we assume the data is normally distributed. The mean zero assumption is done simply so that we can focus on the more interesting and challenging issue of covariance estimation. The Gaussianity assumption can also be relaxed considerably, but for the sake of simplicity we will assume the data is normally distributed.

We wish to understand the behavior of the sample covariance

$$\widehat{\mathbf{\Sigma}}_n = \frac{1}{n} \sum_{j=1}^{n} X_j X_j^T \tag{3}$$

in the setting where $p, n \to \infty$ and $p/n \to \gamma \in (0, \infty)$. First, we observe that in any setting (not just in high dimensions), the eigenvectors of $\widehat{\mathbf{\Sigma}}_n$ are uniformly random over the orthogonal group $\mathcal{O}(p)$ of $\mathbb{R}^p$:

Proposition 1.1. *Let $X_1, \dots, X_n$ be iid $\mathcal{N}(\mathbf{0}, \mathbf{I}_p)$, with sample covariance $\widehat{\mathbf{\Sigma}}_n = \frac{1}{n}\sum_{j=1}^n X_j X_j^T$. Then there is an orthogonal matrix $\widehat{\mathbf{U}}$ of eigenvectors of $\widehat{\mathbf{\Sigma}}_n$ such that $\mathbf{V}\widehat{\mathbf{U}}$ has the same distribution as $\widehat{\mathbf{U}}$ for any fixed orthogonal matrix $\mathbf{V}$.*

Proof. Exercise. (Hint: use the orthogonal invariance of the Gaussian distribution.) $\square$

More interestingly, we will examine the behavior of the spectrum of $\widehat{\mathbf{\Sigma}}_n$. If p is held fixed and $n \to \infty$, then $\widehat{\mathbf{\Sigma}}_n \to \mathbf{\Sigma}$ almost surely, and each of the p eigenvalues of $\widehat{\mathbf{\Sigma}}_n$ converges to 1 almost surely. However, when $p, n \to \infty$, an entirely different phenomenon sets in. Let $\lambda_1, \dots, \lambda_p$ denote the eigenvalues of $\widehat{\mathbf{\Sigma}}_n$. We define the *empirical spectral distribution (ESD)* as follows:

$$\widehat{\mu}_n = \frac{1}{p} \sum_{j=1}^{p} \delta_{\lambda_j}. \tag{4}$$

That is, $\widehat{\mu}_n$ is a discrete probability measure that places mass $1/p$ at each of the p eigenvalues λ_j of $\widehat{\mathbf{\Sigma}}_n$. Then the ESD $\widehat{\mu}_n$ converges to the measure

$$d\mu_\gamma(x) = (1 - 1/\gamma)_+ \delta_0(x) + \frac{\sqrt{(x - b_-)(b_+ - x)}}{2\pi\gamma x} \chi_{[b_-, b_+]}(x), \tag{5}$$

where $b_\pm = (1 \pm \sqrt{\gamma})^2$. This measure is called the *Marchenko-Pastur distribution.*

More precisely, we have the following theorem. Recall that a sequence of probability measures μ_n is said to converge to a limiting measure μ *in distribution* if $\mu_n(-\infty, x] \to \mu(-\infty, x]$ at all x where the mapping $y \mapsto \mu(-\infty, y]$ is continuous.

Theorem 1.2 (Marchenko-Pastur)**.** *Suppose $X_1, \dots, X_n$ are iid $\mathcal{N}(\mathbf{0}, \mathbf{I}_p)$ with sample covariance $\widehat{\mathbf{\Sigma}}_n$ and corresponding ESD $\widehat{\mu}_n$. Then almost surely, $\widehat{\mu}_n \to \mu_\gamma$ in distribution as $p, n \to \infty$ with $p/n \to \gamma$.*

The Marchenko-Pastur distribution μ_γ was derived (in more general form) in the paper [3]; see also the reference [4]. The derivation of the Marchenko-Pastur law uses several "key ingredients" that are of general interest, which we now describe.

1.1 Ingredients for the proof

1.1.1 The Stieltjes transform

The first key ingredient of the derivation is the *Stieltjes transform* of a probability measure. If μ is a probability measure on $\mathbb{R}$, its *Stieltjes transform* (also known as the Cauchy transform, or the Cauchy-Stieltjes transform) is defined as:

$$s_\mu(z) = \int_{\mathbb{R}} \frac{1}{\lambda - z} d\mu(\lambda). \tag{6}$$

This is well-defined for any value of z lying outside the support of μ; in particular, any z with non-zero imaginary part.

Recall that a sequence of measures $\mu_n \to \mu$ *vaguely* if $\int f d\mu_n \to \int f d\mu$ for all continuous functions f with compact support; and $\mu_n \to \mu$ *weakly* if $\int f d\mu_n \to \int f d\mu$ for all bounded, continuous functions f. For probability measures, weak convergence is equivalent to convergence in distribution.

Theorem 1.3 (Stieltjes Continuity Theorem)**.** *If $\mu_1, \mu_2, \dots$ are a sequence of probability measures on $\mathbb{R}$ with Stieltjes transforms $s_1, s_2, \dots,$ and $s_n(z) \to s(z)$ for every $z \in \mathbb{C}^+$, then $s(z)$ is the Stieltjes transform of a subprobability measure μ and $\mu_n \to \mu$ vaguely. If μ is also a probability measure, then $\mu_n \to \mu$ weakly.*

For a proof, see Theorem 2.4.4 in [1], and use the fact that if $\mu_n \to \mu$ vaguely and μ is also a probability measure, then $\mu_n \to \mu$ weakly as well.

Theorem 1.4 (Stieltjes Inversion Formula)**.** *The Stieltjes transform determines the measure μ via the formula*

$$\mu([a, b]) = \lim_{y \to 0^+} \frac{1}{\pi} \int_a^b \Im s_\mu(x + iy) dx. \tag{7}$$

Proof. The key observation behind Theorem 1.4 is that if $z = x + iy$, then

$$\begin{aligned}
\frac{1}{\pi} \Im \frac{1}{\lambda - z} &= \frac{1}{\pi} \Im \frac{(\lambda - x) + iy}{(\lambda - x)^2 + y^2} \\
&= \frac{1}{\pi} \frac{y}{(\lambda - x)^2 + y^2} \\
&= P_y(\lambda - x),
\end{aligned} \tag{8}$$

where P_y is the Poisson kernel. Consequently,

$$\begin{aligned}
\frac{1}{\pi} \int_a^b \Im s_\mu(x + iy) dx &= \frac{1}{\pi} \int_a^b \Im \int_{\mathbb{R}} \frac{1}{\lambda - (x + iy)} d\mu(\lambda)\, dx \\
&= \int_a^b \int_{\mathbb{R}} \frac{1}{\pi} \Im \frac{1}{\lambda - (x + iy)} d\mu(\lambda)\, dx \\
&= \int_a^b \int_{\mathbb{R}} P_y(\lambda - x) d\mu(\lambda)\, dx \\
&= (P_y * \mu)([a, b]),
\end{aligned} \tag{9}$$

the measure of the convolution $P_y * \mu$ over $[a, b]$. Since P_y is an approximation to the identity as $y \to 0^+$, this converges to $\mu([a, b])$ as $y \to 0^+$, completing the proof of the inversion formula. □

Lemma 1.5. *The Stieltjes transform $s_\mu(z)$ is an analytic function on $\mathbb{C}^+$.*

Proof. To show analyticity, we use Morera's Theorem and show that the integral of s_μ around any piecewise C^1 closed curve C contained entirely within $\mathbb{C}^+$ must vanish:

$$\int_C s_\mu(z)dz = \int_C \int_{\mathbb{R}} \frac{1}{x-z} d\mu(x)dz = \int_{\mathbb{R}} \int_C \frac{1}{x-z} dz d\mu(x) = 0, \tag{10}$$

since for any $x \in \mathbb{R}$, the function $(x-z)^{-1}$ is analytic in $\mathbb{C}^+$ and so its integral along C vanishes. Note that the interchange of the order of integration is justified by Fubini's Theorem, since the integrand $(x-z)^{-1}$ is absolutely integrable as it is bounded on the domain of integration, and the integration measure is finite. □

Lemma 1.6. *Suppose $\Im(z) > 0$. Then $\Im(s_\mu(z)) > 0$.*

Proof. Let $z = u + iv$, where $v > 0$. Then:

$$\begin{aligned} s_\mu(z) &= \int_{\mathbb{R}} \frac{1}{\lambda - z} d\mu(\lambda) \\ &= \int_{\mathbb{R}} \frac{1}{\lambda - u - iv} d\mu(\lambda) \\ &= \int_{\mathbb{R}} \frac{\lambda - u + iv}{(\lambda-u)^2 + v^2} d\mu(\lambda) \\ &= \int_{\mathbb{R}} \frac{\lambda - u}{(\lambda-u)^2 + v^2} d\mu(\lambda) + i \int_{\mathbb{R}} \frac{v}{(\lambda-u)^2 + v^2} d\mu(\lambda), \end{aligned} \tag{11}$$

which has positive imaginary part since $v > 0$. □

1.1.2 Concentration of quadratic forms

Theorem 1.7 (Hanson-Wright). *Suppose $X \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_p)$, and $\mathbf{A}$ is a p-by-p real, symmetric matrix. Then*

$$\mathbb{P}\left\{|X^T \mathbf{A} X - \mathrm{tr}(\mathbf{A})| \ge t\right\} \le 2 \exp\left\{-C \min\left(\frac{t^2}{\|\mathbf{A}\|_{\mathrm{F}}^2}, \frac{t}{\|\mathbf{A}\|_{\mathrm{op}}}\right)\right\} \tag{12}$$

for some constant $C > 0$ independent of t and p.

There are many versions of the Hanson-Wright inequality, which apply to a much larger class of random vectors than Gaussian and to non-symmetric matrices $\mathbf{A}$; we restrict to this special case for simplicity, and because it is all we need to derive the Marchenko-Pastur formula. We will prove Hanson-Wright after stating some preliminary results. The following gives the moment generating function for a χ_1^2 random variable:

Lemma 1.8. *Let $z \sim \mathcal{N}(0, 1)$, and $b < 1/2$. Then*

$$\mathbb{E}\left[\exp\left\{bz^2\right\}\right] = \frac{1}{\sqrt{1-2b}}. \tag{13}$$

Proof. Let $\sigma^2 = 1/(1-2b)$. Write the expectation as an integral:

$$\begin{aligned} \mathbb{E}\left[\exp\left\{bz^2\right\}\right] &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{bz^2} e^{-z^2/2} dz \\ &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-z^2(1-2b)/2} dz, \\ &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-z^2/2\sigma^2} dz, \\ &= \frac{1}{\sqrt{1-2b}} \frac{1}{\sqrt{2\pi\sigma^2}} \int_{\mathbb{R}} e^{-z^2/2\sigma^2} dz, \\ &= \frac{1}{\sqrt{1-2b}}, \end{aligned} \tag{14}$$

as claimed. □

Corollary 1.9. *Let $z \sim \mathcal{N}(0,1)$, and $b < 1/2$. Then*

$$\mathbb{E}\left[\exp\left\{b(z^2-1)\right\}\right] \le \exp\left\{\frac{4b^2}{1-2b}\right\} \tag{15}$$

Proof. By Lemma 1.8,

$$\mathbb{E}\left[\exp\left\{b(z^2-1)\right\}\right] = \frac{e^{-b}}{\sqrt{1-2b}}. \tag{16}$$

Taking logarithms, we want to show that

$$f(b) \equiv \frac{4b^2}{1-2b} + b + \frac{1}{2}\log(1-2b) \ge 0. \tag{17}$$

Since $f(0) = 0$, it is enough to show that $f'(b) \ge 0$ when $b \ge 0$ and $f'(b) < 0$ when $b < 0$:

$$\begin{aligned} f'(b) &= \frac{(1-2b)(8b)+8b^2}{(1-2b)^2} + 1 - \frac{1}{1-2b} \\ &= \frac{2b(3-2b)}{1-2b} \end{aligned} \tag{18}$$

which is non-negative when $0 \le b < 1/2$ and negative when $b < 0$, completing the proof. □

We also recall Markov's inequality:

Proposition 1.10 (Markov)**.** *If X is a finite mean, non-negative random variable, then for any $t > 0$*

$$\mathbb{P}(X \ge t) \le \frac{\mathbb{E}[X]}{t}. \tag{19}$$

We now prove Hanson-Wright:

Proof of Hanson-Wright. Since the $\mathcal{N}(\mathbf{0}, \mathbf{I}_p)$ distribution is orthogonally invariant, we may work in the orthonormal eigenbasis of $\mathbf{A}$ and assume that $\mathbf{A}$ is diagonal. Let $a_1, a_2, \dots, a_p$ denote the diagonal entries of $\mathbf{A}$. Then

$$X^T \mathbf{A} X - \mathrm{tr}(\mathbf{A}) = \sum_{j=1}^p a_j (x_j^2 - 1). \tag{20}$$

For brevity, let $W = X^T \mathbf{A} X - \mathrm{tr}(\mathbf{A})$. For any λ between 0 and $1/(2\|\mathbf{A}\|_{\mathrm{op}}) = 1/(2\max_j |a_j|)$, we have:

$$\begin{aligned} \mathbb{P}\{W \ge t\} &= \mathbb{P}\{e^{\lambda W} \ge e^{\lambda t}\} \\ &\le e^{-\lambda t} \prod_{j=1}^p \mathbb{E}\left[\exp\left\{\lambda a_j (x_j^2 - 1)\right\}\right] \\ &\le e^{-\lambda t} \prod_{j=1}^p \exp\left\{\frac{4\lambda^2 a_j^2}{1-2\lambda a_j}\right\} \\ &\le e^{-\lambda t} \prod_{j=1}^p \exp\left\{\frac{4\lambda^2 a_j^2}{1-2\lambda \|\mathbf{A}\|_{\mathrm{op}}}\right\} \\ &= e^{-\lambda t} \exp\left\{\sum_{j=1}^p \frac{4\lambda^2 a_j^2}{1-2\lambda \|\mathbf{A}\|_{\mathrm{op}}}\right\} \\ &= e^{-\lambda t} \exp\left\{\frac{4\lambda^2 \|\mathbf{A}\|_{\mathrm{F}}^2}{1-2\lambda \|\mathbf{A}\|_{\mathrm{op}}}\right\}. \end{aligned} \tag{21}$$

By further restricting λ to lie between 0 and $1/(4\|\mathbf{A}\|_{\text{op}})$, we have $1 - 2\lambda\|\mathbf{A}\|_{\text{op}} \geq 1/2$, and we can simplify the bound:

$$\begin{aligned}
\mathbb{P}\{W \geq t\} &\leq e^{-\lambda t} \exp\left\{\frac{4\lambda^2\|\mathbf{A}\|_{\text{F}}^2}{1 - 2\lambda\|\mathbf{A}\|_{\text{op}}}\right\} \\
&\leq e^{-\lambda t} \exp\left\{8\lambda^2\|\mathbf{A}\|_{\text{F}}^2\right\} \\
&= \exp\left\{8\lambda^2\|\mathbf{A}\|_{\text{F}}^2 - \lambda t\right\}. \qquad (22)
\end{aligned}$$

We now choose the value of λ between 0 and $1/(4\|\mathbf{A}\|_{\text{op}})$ that minimizes the quadratic $q(\lambda) = 8\lambda^2\|\mathbf{A}\|_{\text{F}}^2 - \lambda t$. For brevity, let $\lambda_0 = 1/(4\|\mathbf{A}\|_{\text{op}})$. First, the unconstrained minimum of $q(\lambda)$ is $\lambda_1 = t/(16\|\mathbf{A}\|_{\text{F}}^2)$; λ_1 is also the constrained minimum if it is less than λ_0. If $\lambda_0 \leq \lambda_1$, then the constrained minimum occurs at λ_0 itself, the right endpoint of the domain (since the global minimum λ_1 lies to the right of the domain). That is, we set

$$\lambda^* = \begin{cases} t/(16\|\mathbf{A}\|_{\text{F}}^2), & \text{if } t/(16\|\mathbf{A}\|_{\text{F}}^2) < 1/(4\|\mathbf{A}\|_{\text{op}}) \\ 1/(4\|\mathbf{A}\|_{\text{op}}), & \text{if } t/(16\|\mathbf{A}\|_{\text{F}}^2) \geq 1/(4\|\mathbf{A}\|_{\text{op}}) \end{cases}. \qquad (23)$$

In the first case, the quadratic is equal to:

$$\begin{aligned}
q(t/(16\|\mathbf{A}\|_{\text{F}}^2)) &= 8(t/(16\|\mathbf{A}\|_{\text{F}}^2))^2\|\mathbf{A}\|_{\text{F}}^2 - (t/(16\|\mathbf{A}\|_{\text{F}}^2))t \\
&= \frac{t^2}{32\|\mathbf{A}\|_{\text{F}}^2} - \frac{t^2}{16\|\mathbf{A}\|_{\text{F}}^2} \\
&= -\frac{t^2}{16\|\mathbf{A}\|_{\text{F}}^2}. \qquad (24)
\end{aligned}$$

In the second case, the quadratic is equal to:

$$\begin{aligned}
q(1/(4\|\mathbf{A}\|_{\text{op}})) &= 8(1/(4\|\mathbf{A}\|_{\text{op}}))^2\|\mathbf{A}\|_{\text{F}}^2 - (1/(4\|\mathbf{A}\|_{\text{op}}))t \\
&= \frac{\|\mathbf{A}\|_{\text{F}}^2}{2\|\mathbf{A}\|_{\text{op}}^2} - \frac{t}{4\|\mathbf{A}\|_{\text{op}}}, \qquad (25)
\end{aligned}$$

and because in this case we have the inequality $\|\mathbf{A}\|_{\text{F}}^2 \leq t\|\mathbf{A}\|_{\text{op}}/4$, it follows that

$$\begin{aligned}
q(1/(4\|\mathbf{A}\|_{\text{op}})) &= \frac{\|\mathbf{A}\|_{\text{F}}^2}{2\|\mathbf{A}\|_{\text{op}}^2} - \frac{t}{4\|\mathbf{A}\|_{\text{op}}} \\
&\leq \frac{t\|\mathbf{A}\|_{\text{op}}}{8\|\mathbf{A}\|_{\text{op}}^2} - \frac{t}{4\|\mathbf{A}\|_{\text{op}}} \\
&= \frac{t}{8\|\mathbf{A}\|_{\text{op}}} - \frac{t}{4\|\mathbf{A}\|_{\text{op}}} \\
&= -\frac{t}{8\|\mathbf{A}\|_{\text{op}}}. \qquad (26)
\end{aligned}$$

Therefore, we have the bound

$$q(\lambda^*) \leq -C \min\left(\frac{t^2}{\|\mathbf{A}\|_{\text{F}}^2}, \frac{t}{\|\mathbf{A}\|_{\text{op}}}\right), \qquad (27)$$

for a positive constant C, which shows

$$\mathbb{P}\{W \geq t\} \leq \exp\left\{-C \min\left(\frac{t^2}{\|\mathbf{A}\|_{\text{F}}}, \frac{t}{\|\mathbf{A}\|_{\text{op}}}\right)\right\}. \qquad (28)$$

Repeating the same argument with $-\mathbf{A}$ in place of $\mathbf{A}$ and using the union bound, we have shown

$$\mathbb{P}\{|X^T\mathbf{A}X - \text{tr}(\mathbf{A})| \geq t\} \leq 2\exp\left\{-C \min\left(\frac{t^2}{\|\mathbf{A}\|_{\text{F}}}, \frac{t}{\|\mathbf{A}\|_{\text{op}}}\right)\right\}, \qquad (29)$$

which is the desired bound.

$\square$

Corollary 1.11. *Suppose $X \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_p)$, and $\mathbf{A}$ is a p-by-p complex, symmetric matrix. Then*

$$\mathbb{P}\left\{|X^T \mathbf{A} X - \operatorname{tr}(\mathbf{A})| \geq t\right\} \leq 4 \exp\left\{-C \min\left(\frac{t^2}{\|\mathbf{A}\|_{\mathrm{F}}^2}, \frac{t}{\|\mathbf{A}\|_{\mathrm{op}}}\right)\right\} \tag{30}$$

for some constant $C > 0$ independent of t and p.

Proof. Let $\mathbf{A} = \mathbf{B} + i\mathbf{C}$, where $\mathbf{B}$ and $\mathbf{C}$ are both real and symmetric. Then $\|\mathbf{A}\|_{\mathrm{F}}^2 = \|\mathbf{B}\|_{\mathrm{F}}^2 + \|\mathbf{C}\|_{\mathrm{F}}^2 \geq \|\mathbf{B}\|_{\mathrm{F}}^2$; and for all $\mathbf{x} \in \mathbb{R}^p$,

$$\|\mathbf{A}\mathbf{x}\|^2 = \|\mathbf{B}\mathbf{x}\|^2 + \|\mathbf{C}\mathbf{x}\|^2 \geq \|\mathbf{B}\mathbf{x}\|^2; \tag{31}$$

consequently

$$\begin{aligned} \|\mathbf{A}\|_{\mathrm{op}} &= \sup_{\substack{\mathbf{z} \in \mathbb{C}^p: \\ \|\mathbf{z}\|=1}} \|\mathbf{A}\mathbf{z}\| \\ &\geq \sup_{\substack{\mathbf{x} \in \mathbb{R}^p: \\ \|\mathbf{x}\|=1}} \|\mathbf{A}\mathbf{x}\| \\ &\geq \sup_{\substack{\mathbf{x} \in \mathbb{R}^p: \\ \|\mathbf{x}\|=1}} \|\mathbf{B}\mathbf{x}\| \\ &= \|\mathbf{B}\|_{\mathrm{op}}. \end{aligned} \tag{32}$$

Consequently, by Hanson-Wright for real matrices,

$$\begin{aligned} \mathbb{P}\left\{|X^T \mathbf{B} X - \operatorname{tr}(\mathbf{B})| \geq t/2\right\} &\leq 2 \exp\left\{-C \min\left(\frac{t^2}{\|\mathbf{B}\|_{\mathrm{F}}^2}, \frac{t}{\|\mathbf{B}\|_{\mathrm{op}}}\right)\right\} \\ &\leq 2 \exp\left\{-C \min\left(\frac{t^2}{\|\mathbf{A}\|_{\mathrm{F}}^2}, \frac{t}{\|\mathbf{A}\|_{\mathrm{op}}}\right)\right\}. \end{aligned} \tag{33}$$

Similarly,

$$\mathbb{P}\left\{|X^T \mathbf{C} X - \operatorname{tr}(\mathbf{C})| \geq t/2\right\} \leq 2 \exp\left\{-C \min\left(\frac{t^2}{\|\mathbf{A}\|_{\mathrm{F}}^2}, \frac{t}{\|\mathbf{A}\|_{\mathrm{op}}}\right)\right\}. \tag{34}$$

Consequently, from the union bound

$$\begin{aligned} \mathbb{P}\left\{|X^T \mathbf{A} X - \operatorname{tr}(\mathbf{A})| \geq t\right\} &\leq \mathbb{P}\left\{|X^T \mathbf{B} X - \operatorname{tr}(\mathbf{B})| \geq t/2\right\} + \mathbb{P}\left\{|X^T \mathbf{C} X - \operatorname{tr}(\mathbf{C})| \geq t/2\right\} \\ &\leq 4 \exp\left\{-C \min\left(\frac{t^2}{\|\mathbf{A}\|_{\mathrm{F}}^2}, \frac{t}{\|\mathbf{A}\|_{\mathrm{op}}}\right)\right\}, \end{aligned} \tag{35}$$

as desired. $\square$

Corollary 1.12. *Suppose $X \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_p)$, and $\mathbf{A}_p$ is a sequence of p-by-p complex symmetric matrices with $\|\mathbf{A}_p\|_{\mathrm{op}} \leq K$ for some constant $K > 0$. Then*

$$\frac{|X^T \mathbf{A}_p X - \operatorname{tr}(\mathbf{A}_p)|}{p} \to 0 \tag{36}$$

almost surely as $p \to \infty$.

Before proving Corollary 1.12, we recall a version of the Borel-Cantelli Lemma from probability:

Lemma 1.13 (Borel-Cantelli)**.** *Let $A_1, A_2, \dots$ be a sequence of random numbers. If for every choice of sufficiently small $t > 0$ we have*

$$\sum_{n=1}^{\infty} \mathbb{P}(|A_n| > t) < \infty, \tag{37}$$

then $A_n \to 0$ almost surely.

We now prove Corollary 1.12:

Proof of Corollary 1.12. Let $\mathbf{A} \equiv \mathbf{A}_p$, to ease the notation. Fix $t > 0$. Because of the inequality $\|\mathbf{A}\|_{\mathrm{F}}^2 \le p\|\mathbf{A}\|_{\mathrm{op}}^2 \le K^2 p$, by Hanson-Wright,

$$\begin{aligned}
\mathbb{P}\left\{\frac{|X^T\mathbf{A}X - \mathrm{tr}(\mathbf{A})|}{p} \ge t\right\} &\le 4\exp\left\{-C\min\left(\frac{p^2t^2}{\|\mathbf{A}\|_{\mathrm{F}}^2}, \frac{pt}{\|\mathbf{A}\|_{\mathrm{op}}}\right)\right\} \\
&\le 4\exp\left\{-C\min\left(\frac{p^2t^2}{K^2p}, \frac{pt}{K}\right)\right\} \\
&\le 4\exp\left\{-C\min\left(\frac{pt^2}{K^2}, \frac{pt}{K}\right)\right\}. \qquad (38)
\end{aligned}$$

Consequently, for all small $t > 0$,

$$\sum_{p=1}^{\infty}\mathbb{P}\left\{\frac{|X^T\mathbf{A}X - \mathrm{tr}(\mathbf{A})|}{p} \ge t\right\} < \infty, \qquad (39)$$

and the result now follows by Borel-Cantelli. □

1.1.3 The Sherman-Morrison formula

At the heart of the proof of Marchenko-Pastur is the following formula from linear algebra.

Proposition 1.14 (Sherman-Morrison)**.** *Let $\mathbf{C}$ be an invertible matrix of size $p \times p$, and $\mathbf{a}$ be any vector in $\mathbb{R}^p$. Then $\mathbf{C}+\mathbf{a}\mathbf{a}^T$ is invertible if and only if $\mathbf{a}^T\mathbf{C}^{-1}\mathbf{a} \neq -1$, and*

$$(\mathbf{C}+\mathbf{a}\mathbf{a}^T)^{-1} = \mathbf{C}^{-1} - \frac{\mathbf{C}^{-1}\mathbf{a}\mathbf{a}^T\mathbf{C}^{-1}}{1+\mathbf{a}^T\mathbf{C}^{-1}\mathbf{a}}. \qquad (40)$$

If we assume $\mathbf{a}^T\mathbf{C}^{-1}\mathbf{a} \neq -1$, the formula for $(\mathbf{C}+\mathbf{a}\mathbf{a}^T)^{-1}$ can be easily verified by multiplying the right side by $\mathbf{C}+\mathbf{a}\mathbf{a}^T$ and simplifying the resulting expression via straightforward algebra. In fact, this argument proves that the condition $\mathbf{a}^T\mathbf{C}^{-1}\mathbf{a} \neq -1$ is sufficient for the existence of $(\mathbf{C}+\mathbf{a}\mathbf{a}^T)^{-1}$.

Corollary 1.15. *If $\mathbf{C}$ is an invertible matrix of size $p \times p$, and $\mathbf{a}$ is any vector in $\mathbb{R}^p$ such that $\mathbf{C}+\mathbf{a}\mathbf{a}^T$ is invertible, then $\mathbf{a}^T\mathbf{C}^{-1}\mathbf{a} \neq -1$ and*

$$\mathbf{a}^T(\mathbf{C}+\mathbf{a}\mathbf{a}^T)^{-1}\mathbf{a} = \frac{\mathbf{a}^T\mathbf{C}^{-1}\mathbf{a}}{1+\mathbf{a}^T\mathbf{C}^{-1}\mathbf{a}}. \qquad (41)$$

Proof. The proof follows straightforwardly from Sherman-Morrison:

$$\begin{aligned}
\mathbf{a}^T(\mathbf{C}+\mathbf{a}\mathbf{a}^T)^{-1}\mathbf{a} &= \mathbf{a}^T\mathbf{C}^{-1}\mathbf{a} - \frac{\mathbf{a}^T\mathbf{C}^{-1}\mathbf{a}\mathbf{a}^T\mathbf{C}^{-1}\mathbf{a}}{1+\mathbf{a}^T\mathbf{C}^{-1}\mathbf{a}} \\
&= \frac{\mathbf{a}^T\mathbf{C}^{-1}\mathbf{a}(1+\mathbf{a}^T\mathbf{C}^{-1}\mathbf{a}) - \mathbf{a}^T\mathbf{C}^{-1}\mathbf{a}\mathbf{a}^T\mathbf{C}^{-1}\mathbf{a}}{1+\mathbf{a}^T\mathbf{C}^{-1}\mathbf{a}} \\
&= \frac{\mathbf{a}^T\mathbf{C}^{-1}\mathbf{a} + \mathbf{a}^T\mathbf{C}^{-1}\mathbf{a}\mathbf{a}^T\mathbf{C}^{-1}\mathbf{a} - \mathbf{a}^T\mathbf{C}^{-1}\mathbf{a}\mathbf{a}^T\mathbf{C}^{-1}\mathbf{a}}{1+\mathbf{a}^T\mathbf{C}^{-1}\mathbf{a}} \\
&= \frac{\mathbf{a}^T\mathbf{C}^{-1}\mathbf{a}}{1+\mathbf{a}^T\mathbf{C}^{-1}\mathbf{a}}, \qquad (42)
\end{aligned}$$

as claimed. □

We will also need the following result, which also follows from the Sherman-Morrison formula.

Lemma 1.16. *Suppose $\mathbf{A}$ is a real, symmetric, p-by-p matrix, $\mathbf{x}$ is a vector in $\mathbb{R}^p$, and $z = u + iv$ with $v > 0$. Then*

$$\left|\mathrm{tr}[(\mathbf{A}+\mathbf{x}\mathbf{x}^T - z\mathbf{I}_p)^{-1}] - \mathrm{tr}[(\mathbf{A} - z\mathbf{I}_p)^{-1}]\right| \le \frac{1}{v}. \qquad (43)$$

Proof. If $\mathbf{A} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^T$ is the eigenvalue decomposition of $\mathbf{A}$, then by replacing $\mathbf{x}$ by $\mathbf{U}^T\mathbf{x}$, we can assume without loss of generality that $\mathbf{A} = \mathrm{diag}(a_1, \dots, a_p)$ is diagonal. By the Sherman-Morrison formula,

$$(\mathbf{A} - z\mathbf{I}_p)^{-1} - (\mathbf{A} + \mathbf{x}\mathbf{x}^T - z\mathbf{I}_p)^{-1} = \frac{(\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x}\mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-1}}{1 + \mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x}}. \tag{44}$$

Taking traces, we get:

$$\mathrm{tr}[(\mathbf{A} - z\mathbf{I}_p)^{-1}] - \mathrm{tr}[(\mathbf{A} + \mathbf{x}\mathbf{x}^T - z\mathbf{I}_p)^{-1}] = \frac{\mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-2}\mathbf{x}}{1 + \mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x}}. \tag{45}$$

The numerator of the right side may be bounded as follows:

$$\begin{aligned}
\mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-2}\mathbf{x} &= \langle (\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x}, (\mathbf{A} - \overline{z}\mathbf{I}_p)^{-1}\mathbf{x} \rangle_{\mathbb{C}^p} \\
&\le \|(\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x}\|^2 \\
&= \sum_{j=1}^p \frac{x_j^2}{|a_j - z|^2}.
\end{aligned} \tag{46}$$

As for the denominator, since $\Im(1 + \mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x}) = \Im(\mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x})$, we write:

$$\begin{aligned}
|1 + \mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x}| &\ge |\Im(\mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x})| \\
&= \left| \Im \sum_{j=1}^p \frac{x_j^2}{a_j - z} \right| \\
&= \left| \Im \sum_{j=1}^p \frac{x_j^2(a_j - \overline{z})}{|a_j - z|^2} \right| \\
&= \left| \Im \sum_{j=1}^p \frac{x_j^2(a_j - u + iv)}{|a_j - z|^2} \right| \\
&= \left| \sum_{j=1}^p \frac{x_j^2 v}{|a_j - z|^2} \right| \\
&= v \sum_{j=1}^p \frac{x_j^2}{|a_j - z|^2},
\end{aligned} \tag{47}$$

where the last equality holds because $v > 0$. Consequently,

$$\begin{aligned}
\left| \mathrm{tr}[(\mathbf{A} + \mathbf{x}\mathbf{x}^T - z\mathbf{I}_p)^{-1}] - \mathrm{tr}[(\mathbf{A} - z\mathbf{I}_p)^{-1}] \right| &= \left| \frac{\mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-2}\mathbf{x}}{1 + \mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x}} \right| \\
&\le \frac{\sum_{j=1}^p \frac{x_j^2}{|a_j - z|^2}}{v \sum_{j=1}^p \frac{x_j^2}{|a_j - z|^2}} \\
&= \frac{1}{v},
\end{aligned} \tag{48}$$

as desired. □

1.1.4 Additional lemmas

Lemma 1.17. *Suppose $z = u + iv$ with $v > 0$, $\mathbf{A}$ is a real, symmetric, positive semi-definite p-by-p matrix, and $\mathbf{x} \in \mathbb{R}^p$. Then*

$$\Im(z + z\mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x}) \ge v. \tag{49}$$

Proof. Since $\Im(z + z\mathbf{x}^T(\mathbf{A} - z)\mathbf{I}_p^{-1}\mathbf{x}) = v + \Im(z\mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x})$, we must show that

$$\Im(z\mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x}) \geq 0. \tag{50}$$

Since $\mathbf{x}$ is arbitrary, by changing into the eigenbasis of $\mathbf{A}$ we may assume without loss of generality that $\mathbf{A} = \text{diag}(a_1, \dots, a_p)$, with each $a_j \geq 0$. We then have

$$\begin{aligned} z\mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x} &= \sum_{j=1}^p \frac{z}{a_j - z} x_j^2 \\ &= \sum_{j=1}^p \frac{u + iv}{a_j - u - iv} x_j^2 \\ &= \sum_{j=1}^p \frac{(u + iv)(a_j - u + iv)}{(a_j - u)^2 + v^2} x_j^2 \\ &= \sum_{j=1}^p \frac{u(a_j - u) - v^2}{(a_j - u)^2 + v^2} x_j^2 + i \sum_{j=1}^p \frac{va_j}{(a_j - u)^2 + v^2} x_j^2, \end{aligned} \tag{51}$$

whose imaginary part is

$$\Im(z\mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x}) = \sum_{j=1}^p \frac{va_j}{(a_j - u)^2 + v^2} x_j^2 \geq 0, \tag{52}$$

since $a_j \geq 0$ and $v > 0$. □

Corollary 1.18. *Suppose $z = u + iv$ with $v > 0$, $\mathbf{A}$ is a real, symmetric, positive semi-definite p-by-p matrix, and $\mathbf{x} \in \mathbb{R}^p$. Then*

$$\frac{1}{|1 + \mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x}|} \leq \frac{|z|}{v}. \tag{53}$$

Proof. Since $\Im(ab) \leq |a||b|$ for any complex a and b, in particular

$$\begin{aligned} |\Im(z + z\mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x})| &= |\Im(z(1 + \mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x}))| \\ &\leq |z||1 + \mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x}|. \end{aligned} \tag{54}$$

Consequently,

$$\begin{aligned} \frac{1}{|1 + \mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x}|} &\leq \frac{|z|}{|\Im(z + z\mathbf{x}^T(\mathbf{A} - z\mathbf{I}_p)^{-1}\mathbf{x})|} \\ &\leq \frac{|z|}{v}, \end{aligned} \tag{55}$$

where the last inequality follows from Lemma 1.17. □

The next result follows from Lemma 1.17 and Corollary 1.18:

Lemma 1.19. *Suppose $z = u + iv$ with $v > 0$, and $\mathbf{A}$ is a real, symmetric, positive semi-definite p-by-p matrix. Then*

$$\Im(z + z\text{tr}[(\mathbf{A} - z\mathbf{I}_p)^{-1}]) \geq v, \tag{56}$$

and

$$\frac{1}{|1 + \text{tr}[(\mathbf{A} - z\mathbf{I}_p)^{-1}]|} \leq \frac{|z|}{v}. \tag{57}$$

Proof. Without loss of generality, we may suppose $\mathbf{A}$ is diagonal; we may then apply Lemma 1.17 and Corollary 1.18 with $\mathbf{x} = \mathbf{1}$. □

1.2 Derivation of Marchenko-Pastur

With all the key ingredients assembled (the Stieltjes transform, Hanson-Wright, and Sherman-Morrison), we can proceed to the derivation of the Marchenko-Pastur law. The strategy of the proof is derived from the paper [5], though this paper shows the result for a much broader class of random matrices.

Let $\mathbf{X} = [X_1, \dots, X_n]$ denote the random matrix with Gaussian entries of variance 1 and mean 0. The sample covariance is then $\mathbf{A}/n$, where $\mathbf{A} = \mathbf{X}\mathbf{X}^T$ is known as the *scatter matrix* (the unnormalized covariance).

For brevity, denote by $s_n(z) = s_{\widehat{\mu}_n}(z)$ the Stieltjes transform of the ESD. It is given by the following formula:

Lemma 1.20. $s_n(z) = \frac{n}{p}\mathrm{tr}[(\mathbf{A} - zn\mathbf{I}_p)^{-1}]$.

Proof. By definition,

$$\widehat{\mu}_n = \frac{1}{p}\sum_{j=1}^{p} \delta_{\lambda_j}, \tag{58}$$

where $\lambda_1, \dots, \lambda_p$ are the eigenvalues of $\mathbf{A}/n$. The Stieltjes transform is then

$$\begin{aligned} s_n(z) &= \int_{\mathbb{R}} \frac{1}{\lambda - z} d\widehat{\mu}_n(\lambda) \\ &= \frac{1}{p}\sum_{j=1}^{p} \frac{1}{\lambda_j - z} \\ &= \frac{n}{p}\sum_{j=1}^{p} \frac{1}{n\lambda_j - zn} \\ &= \frac{n}{p}\mathrm{tr}[(\mathbf{A} - zn\mathbf{I}_p)^{-1}], \end{aligned} \tag{59}$$

as claimed. □

The main trick in the proof is to augment $\mathbf{X}$ by a single column $X_{n+1} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_p)$ that is independent of $X_1, \dots, X_n$; that is, consider the matrix

$$\widetilde{\mathbf{X}} = [\mathbf{X}|X_{n+1}] = [X_1, \dots, X_n, X_{n+1}]. \tag{60}$$

Define $\widetilde{\mathbf{A}} = \widetilde{\mathbf{X}}\widetilde{\mathbf{X}}^T$. Let $\mathbf{A}_j$ be the p-by-n matrix whose columns are those of $\widetilde{\mathbf{A}}$ but with the j-th column removed; note that $\mathbf{A} = \mathbf{A}_{n+1}$ in this notation. Let $s_{j,n}$ denote the Stieltjes transform of the ESD of $\mathbf{A}_j$. Then by Lemma 1.20,

$$s_{j,n}(z) = \frac{n}{p}\mathrm{tr}[(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}]. \tag{61}$$

Step 1: Derive the matrix equation using Sherman-Morrison. Using Sherman-Morrison (more specifically, Corollary 1.15), we can write:

$$X_j^T(\widetilde{\mathbf{A}} - zn\mathbf{I}_p)^{-1}X_j = X_j^T(\mathbf{A}_j - zn\mathbf{I}_p + X_jX_j^T)^{-1}X_j = \frac{X_j^T(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}X_j}{1 + X_j^T(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}X_j}. \tag{62}$$

Averaging over $j = 1, \dots, n+1$, we find:

$$\frac{1}{n+1}\sum_{j=1}^{n+1} X_j^T(\widetilde{\mathbf{A}} - zn\mathbf{I}_p)^{-1}X_j = \frac{1}{n+1}\sum_{j=1}^{n+1} \frac{X_j^T(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}X_j}{1 + X_j^T(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}X_j}. \tag{63}$$

We rewrite the left side of (63) as follows:

$$\mathbf{I}_p = (\widetilde{\mathbf{A}} - zn\mathbf{I}_p)(\widetilde{\mathbf{A}} - zn\mathbf{I}_p)^{-1} = \sum_{j=1}^{n+1} X_jX_j^T(\widetilde{\mathbf{A}} - zn\mathbf{I}_p)^{-1} - zn(\widetilde{\mathbf{A}} - zn\mathbf{I}_p)^{-1}, \tag{64}$$

and so by taking the trace of each side and dividing by $n+1$, we find:

$$\frac{p}{n+1} = \frac{1}{n+1}\sum_{j=1}^{n+1} X_j^T(\widetilde{\mathbf{A}} - zn\mathbf{I}_p)^{-1}X_j - z\frac{n}{n+1}\mathrm{tr}(\widetilde{\mathbf{A}} - zn\mathbf{I}_p)^{-1} \tag{65}$$

or equivalently

$$\frac{1}{n+1}\sum_{j=1}^{n+1} X_j^T(\widetilde{\mathbf{A}} - zn\mathbf{I}_p)^{-1}X_j = \frac{p}{n+1} + z\frac{n}{n+1}\mathrm{tr}(\widetilde{\mathbf{A}} - zn\mathbf{I}_p)^{-1}. \tag{66}$$

Combining equations (63) and (66) yields:

$$\frac{p}{n+1} + z\frac{n}{n+1}\mathrm{tr}[(\widetilde{\mathbf{A}} - zn\mathbf{I}_p)^{-1}] = \frac{1}{n+1}\sum_{j=1}^{n+1}\frac{X_j^T(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}X_j}{1 + X_j^T(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}X_j}. \tag{67}$$

Step 2: Take the limit using concentration of quadratic forms. Next, we show that equation (67) converges to a simple quadratic equation in the limit $p, n \to \infty$. We start with the following result:

Lemma 1.21. *For $z \in \mathbb{C}^+$ and $1 \le j \le n+1$,*

$$\left|(p/n)s_{j,n}(z) - \mathrm{tr}[(\widetilde{\mathbf{A}} - zn\mathbf{I}_p)^{-1}]\right| \le \frac{1}{n\Im(z)}. \tag{68}$$

Proof. By Lemma 1.16,

$$\begin{aligned}\left|(p/n)s_{j,n}(z) - \mathrm{tr}[(\widetilde{\mathbf{A}} - zn\mathbf{I}_p)^{-1}]\right| &= \left|\mathrm{tr}[(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}] - \mathrm{tr}[(\widetilde{\mathbf{A}} - zn\mathbf{I}_p)^{-1}]\right| \\ &= \left|\mathrm{tr}[(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}] - \mathrm{tr}[(\mathbf{A}_j + X_jX_j^T - zn\mathbf{I}_p)^{-1}]\right| \\ &\le \frac{1}{n\Im(z)} \end{aligned} \tag{69}$$

which is the desired bound. □

Consider the right side of equation (67), and specifically each term of the form $X_j^T(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}X_j$. The matrices $\mathbf{A}_j$ and X_j are independent, since $\mathbf{A}_j$ is defined by removing the j-th column X_j from $\widetilde{\mathbf{A}}$.

The following lemma will help us simplify the right side of equation (67):

Lemma 1.22. *The following limit holds almost surely:*

$$\lim_{n\to\infty}\frac{1}{n+1}\sum_{j=1}^{n+1}\left(\frac{X_j^T(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}X_j}{1 + X_j^T(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}X_j} - \frac{(p/n)s_{j,n}(z)}{1 + (p/n)s_{j,n}(z)}\right) = 0. \tag{70}$$

Proof. For brevity, let $W_{j,n} = X_j^T(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}X_j$. Then

$$\begin{aligned}\frac{X_j^T(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}X_j}{1 + X_j^T(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}X_j} - \frac{(p/n)s_{j,n}(z)}{1 + (p/n)s_{j,n}(z)} &= \frac{W_{j,n}}{1 + W_{j,n}} - \frac{(p/n)s_{j,n}(z)}{1 + (p/n)s_{j,n}(z)} \\ &= \frac{W_{j,n} - (p/n)s_{j,n}(z)}{(1 + W_{j,n})(1 + (p/n)s_{j,n}(z))}. \end{aligned} \tag{71}$$

Now by Corollary 1.18,

$$\begin{aligned}\frac{1}{|1 + W_{j,n}|} &= \frac{1}{|1 + X_j^T(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}X_j|} \\ &\le \frac{|zn|}{n\Im(z)} \\ &\le \frac{|z|}{\Im(z)}; \end{aligned} \tag{72}$$

and by the second part of Lemma 1.19,

$$\begin{aligned}\frac{1}{|1+(p/n)s_{j,n}(z)|} &= \frac{1}{|1+\mathrm{tr}[(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}]|} \\ &\leq \frac{|zn|}{n\Im(z)} \\ &\leq \frac{|z|}{\Im(z)}; \end{aligned} \tag{73}$$

Consequently, continuing (71)

$$\begin{aligned}\left|\frac{X_j^T(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}X_j}{1+X_j^T(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}X_j} - \frac{(p/n)s_{j,n}(z)}{1+(p/n)s_{j,n}(z)}\right| &= \left|\frac{W_{j,n}-(p/n)s_{j,n}(z)}{(1+W_{j,n})(1+(p/n)s_{j,n}(z))}\right| \\ &\leq \frac{|z|^2}{\Im(z)^2}|W_{j,n}-(p/n)s_{j,n}(z)|,\end{aligned} \tag{74}$$

and so

$$\left|\frac{1}{n+1}\sum_{j=1}^{n+1}\left(\frac{X_j^T(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}X_j}{1+X_j^T(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}X_j} - \frac{(p/n)s_{j,n}(z)}{1+(p/n)s_{j,n}(z)}\right)\right| \leq \frac{|z|^2}{\Im(z)^2}\frac{1}{n+1}\sum_{j=1}^{n+1}|W_{j,n}-(p/n)s_{j,n}(z)|. \tag{75}$$

Since the spectrum of $\mathbf{A}_j/n$ is real and z has positive imaginary part, $\|(\mathbf{A}_j/n - z\mathbf{I}_p)^{-1}\|_{\mathrm{op}} \leq K$, for some $K>0$ that is independent of $\mathbf{A}_j$. Furthermore, $\|(\mathbf{A}_j/n - z\mathbf{I}_p)^{-1}\|_{\mathrm{F}}^2 \leq p\|(\mathbf{A}_j/n - z\mathbf{I}_p)^{-1}\|_{\mathrm{op}}^2 \leq K^2 p$.

Let $t>0$. By the union bound and Hanson-Wright,

$$\begin{aligned}\mathbb{P}\left\{\frac{1}{n+1}\sum_{j=1}^{n+1}|W_{j,n}-(p/n)s_{j,n}(z)| \geq t\right\} &\leq \sum_{j=1}^{n+1}\mathbb{P}\left\{|W_{j,n}-(p/n)s_{j,n}(z)|\geq t\right\} \\ &= \sum_{j=1}^{n+1}\mathbb{P}\left\{|X_j^T(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}X_j - \mathrm{tr}[(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}]| \geq t\right\} \\ &= \sum_{j=1}^{n+1}\mathbb{P}\left\{|X_j^T(\mathbf{A}_j/n - z\mathbf{I}_p)^{-1}X_j - \mathrm{tr}[(\mathbf{A}_j/n - z\mathbf{I}_p)^{-1}]| \geq nt\right\} \\ &\leq 4(n+1)\exp\left\{-C\min\left(\frac{n^2t^2}{K^2p},\frac{nt}{K}\right)\right\}\end{aligned} \tag{76}$$

Since $p/n \to \gamma \in (0,\infty)$, it follows that for all $t>0$,

$$\sum_{p=1}^{\infty}\mathbb{P}\left\{\frac{1}{n+1}\sum_{j=1}^{n+1}|W_{j,n}-(p/n)s_{j,n}(z)| \geq t\right\} < \infty, \tag{77}$$

and the result now follows by Borel-Cantelli. □

Corollary 1.23. *The following limit holds almost surely:*

$$\lim_{n\to\infty}\left(\frac{1}{n+1}\sum_{j=1}^{n+1}\frac{X_j^T(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}X_j}{1+X_j^T(\mathbf{A}_j - zn\mathbf{I}_p)^{-1}X_j} - \frac{(p/n)s_n(z)}{1+(p/n)s_n(z)}\right) = 0. \tag{78}$$

Proof. The result will follow from Lemma 1.22 if we can show that

$$\frac{1}{n+1}\sum_{j=1}^{n+1}\frac{(p/n)s_{j,n}(z)}{1+(p/n)s_{j,n}(z)} - \frac{(p/n)s_n(z)}{1+(p/n)s_n(z)} \tag{79}$$

converges to 0 almost surely as $n \to \infty$. For each j, we have

$$\begin{aligned}\left|\frac{(p/n)s_{j,n}(z)}{1+(p/n)s_{j,n}(z)} - \frac{(p/n)s_n(z)}{1+(p/n)s_n(z)}\right| &= \left|\frac{(p/n)s_{j,n}(z) - (p/n)s_n(z)}{(1+(p/n)s_{j,n}(z))(1+(p/n)s_n(z))}\right| \\ &\le \frac{|z|^2}{\Im(z)^2}|(p/n)s_{j,n}(z) - (p/n)s_n(z)|, \end{aligned} \tag{80}$$

where we have used the second part of Lemma 1.19 for the inequality. We have:

$$|(p/n)s_{j,n}(z) - (p/n)s_n(z)| \le \left|(p/n)s_{j,n}(z) - \mathrm{tr}[(\widetilde{\mathbf{A}} - zn\mathbf{I}_p)^{-1}]\right| + \left|(p/n)s_n(z) - \mathrm{tr}[(\widetilde{\mathbf{A}} - zn\mathbf{I}_p)^{-1}]\right|. \tag{81}$$

By Lemma 1.21, we have

$$\left|(p/n)s_{j,n}(z) - \mathrm{tr}[(\widetilde{\mathbf{A}} - zn\mathbf{I}_p)^{-1}]\right| \le \frac{1}{n\Im(z)} \tag{82}$$

and

$$\left|(p/n)s_n(z) - \mathrm{tr}[(\widetilde{\mathbf{A}} - zn\mathbf{I}_p)^{-1}]\right| \le \frac{1}{n\Im(z)} \tag{83}$$

Combining these bounds with (81), we get

$$|(p/n)s_{j,n}(z) - (p/n)s_n(z)| \le \frac{2}{n\Im(z)}, \tag{84}$$

and combining this with (80) gives

$$\begin{aligned}\left|\frac{1}{n+1}\sum_{j=1}^{n+1}\frac{(p/n)s_{j,n}(z)}{1+(p/n)s_{j,n}(z)} - \frac{(p/n)s_n(z)}{1+(p/n)s_n(z)}\right| &\le \frac{1}{n+1}\sum_{j=1}^{n+1}\frac{|z|^2}{\Im(z)^2}|(p/n)s_{j,n}(z) - (p/n)s_n(z)| \\ &\le \frac{1}{n+1}\sum_{j=1}^{n+1}\frac{2|z|^2}{\Im(z)^3}\frac{1}{n} \\ &= \frac{2|z|^2}{\Im(z)^3}\frac{1}{n}, \end{aligned} \tag{85}$$

which proves the result. □

From Corollary 1.23 and $p/n \to \gamma$, it can now be shown that the right side of (67) converges almost surely to

$$\frac{\gamma s_n(z)}{1+\gamma s_n(z)}, \tag{86}$$

as $n \to \infty$. As for the left side, from Lemma 1.21,

$$\begin{aligned}\mathrm{tr}[(\widetilde{\mathbf{A}} - zn\mathbf{I}_p)^{-1}] &\sim \mathrm{tr}[(\mathbf{A} - zn\mathbf{I}_p)^{-1}] \\ &\sim \gamma s_n(z), \end{aligned} \tag{87}$$

where $a_n \sim b_n$ denotes that $a_n - b_n \to 0$ almost surely as $n \to \infty$. Consequently, we arrive at the following asymptotic equation for $s_n(z)$:

$$\gamma + z\gamma s_n(z) \sim \frac{\gamma s_n(z)}{1+\gamma s_n(z)}, \tag{88}$$

or equivalently,

$$\gamma z s_n(z)^2 + (z+\gamma-1)s_n(z) + 1 \sim 0. \tag{89}$$

Since $\Im(z) > 0$, by Lemma 1.6 $\Im(s_n(z)) > 0$ as well. Let $s(z)$ denote the solution with positive imaginary part to the quadratic

$$\gamma z s(z)^2 + (z + \gamma - 1)s(z) + 1 = 0. \tag{90}$$

Since the coefficients of (89) converge almost surely to those of (90), we must then have $s_n(z) \to s(z)$ almost surely.

Step 3: Solve equation (90) for $s(z)$, then derive μ_γ.

We solve equation (90) using the quadratic formula:

$$s(z) = \frac{1 - \gamma - z + \sqrt{(1 - \gamma - z)^2 - 4\gamma z}}{2\gamma z} = \frac{1 - \gamma - z + \sqrt{(z - b_+)(z - b_-)}}{2\gamma z}, \tag{91}$$

where the square root $\sqrt{\cdot}$ is the one with positive imaginary part, as this ensures that $s(z)$ also has positive imaginary part. This can be seen be considering the region where $x \in (b_-, b_+)$, and taking the limit as $y \to 0$; it is straightforward to verify that when y is sufficiently small, the square root with positive imaginary part must be used for $s(z)$ to also lie in $\mathbb{C}^+$. Indeed, if we write

$$\sqrt{(z - b_+)(z - b_-)} = A(z) + iB(z), \tag{92}$$

then when $x \in (b_-, b_+)$, $A(x + iy) \to 0$ as $y \to 0^+$. Consequently, if $z = x + iy$ with $y > 0$,

$$\begin{aligned} s(z) &= \frac{1 - \gamma - z + \sqrt{(z - b_+)(z - b_-)}}{2\gamma z} \\ &= \frac{(1 - \gamma)\overline{z} - |z|^2 + \overline{z}(A(z) + iB(z))}{2\gamma |z|^2} \\ &= \frac{(1 - \gamma)x - (1 - \gamma)iy - |z|^2 + (x - iy)(A(z) + iB(z))}{2\gamma |z|^2} \\ &= \frac{(1 - \gamma)x - (1 - \gamma)iy - |z|^2 + xA(z) + yB(z) - iyA(z) + ixB(z)}{2\gamma |z|^2}, \end{aligned} \tag{93}$$

whose imaginary part is

$$\Im s(z) = \frac{-(1 - \gamma)y - yA(z) + xB(z)}{2\gamma |z|^2}. \tag{94}$$

Since this must be positive, taking the limit as $y \to 0^+$ shows that we must take $B(z) > 0$, as claimed. The fact that this same square root must be used on all of $\mathbb{C}^+$ now follows from the uniqueness of analytic continuations.

Finally, we take the limit of $\Im(s(x + iy))/\pi$ as $y \to 0^+$ to recover the formula for $d\mu_\gamma(x)$, thereby completing the proof. When $x \notin [b_-, b_+] \cup \{0\}$, the limit is easily show to be 0, since $s(x)$ is real for such x; and when $x \in [b_-, b_+]$, it is easily shown to converge to the desired density. If $\gamma \leq 1$, we may show that limit is 0 when $x = 0$ (e.g. by using l'Hopital's rule); indeed, we have

$$\lim_{z \to 0} s(z) = \frac{1}{1 - \gamma}, \tag{95}$$

which is real. The value of the point mass at 0 when $\gamma > 1$ then follows easily from the following lemma, which implies that $\widehat{\mathbf{\Sigma}}_n$ must have $p - n$ eigenvalues 0, with a total mass of $(p - n)/p = 1 - n/p \to 1 - 1/\gamma$:

Lemma 1.24. *Suppose $k < l$, and $\mathbf{G}$ is a k-by-l matrix with iid $\mathcal{N}(0, 1)$ entries. Then with probability 1, $\mathbf{G}$ has rank k.*

Proof. Exercise. □

1.3 Generalizations

It is known that if the eigenvalue distribution of the population covariance $\mathbf{\Sigma}^{(p)}$ (known as the *population spectral distribution*) converges to a limit with measure ν, then the eigenvalue distribution of the sample covariance $\widehat{\mathbf{\Sigma}}_n$ will also converge to a limit, whose Stieltjes transform $s(z)$ satisfies

$$-\frac{1}{\tilde{s}(z)} = z - \gamma \int_{\mathbb{R}} \frac{1}{1 + \lambda \tilde{s}(z)} d\nu(\lambda), \tag{96}$$

where $\tilde{s}(z)$, known as the *companion Stieltjes transform*, is defined by

$$\tilde{s}(z) = \gamma s(z) + \frac{\gamma - 1}{z}. \tag{97}$$

For example, see the paper [4]. While there is typically not a closed form solution for the Stieltjes transform $s(z)$ or the limit of the ESD, $s(z)$ and the ESD may be solved for numerically, and their theoretical properties can be derived from the equation.

These results may also be extended to much more general distributions than just Gaussians. For example, if the vectors X_j have entries with 4 moments (that is, $\mathbb{E}[|X_{ij}|^4] < \infty$), then the ESD converges to the Marchenko-Pastur law. A standard reference with numerous results on sample covariances matrices is the textbook [2].

1.4 Convergence to the edge

The Marchenko-Pastur law applies to the distribution of all the eigenvalues of the sample covariance. A priori, it does not rule out the possibility that a negligibly small fraction of non-zero eigenvalues lie far outside $[b_-, b_+]$ of the Marchenko-Pastur distribution. In fact, it can be shown that this does not occur. More precisely:

Theorem 1.25. *For any fixed index i, $\lambda_i(\widehat{\mathbf{\Sigma}}_n) \to (1 + \sqrt{\gamma})^2$ and $\lambda_{\min\{p,n\}-i}(\widehat{\mathbf{\Sigma}}_n) \to (1 - \sqrt{\gamma})^2$ almost surely as $n \to \infty$.*

We will not prove this result; see [6], or Chapter 5 in [2]. However, the theorem will be very useful for us in understanding the behavior of the signal-plus-noise model in the next section.

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